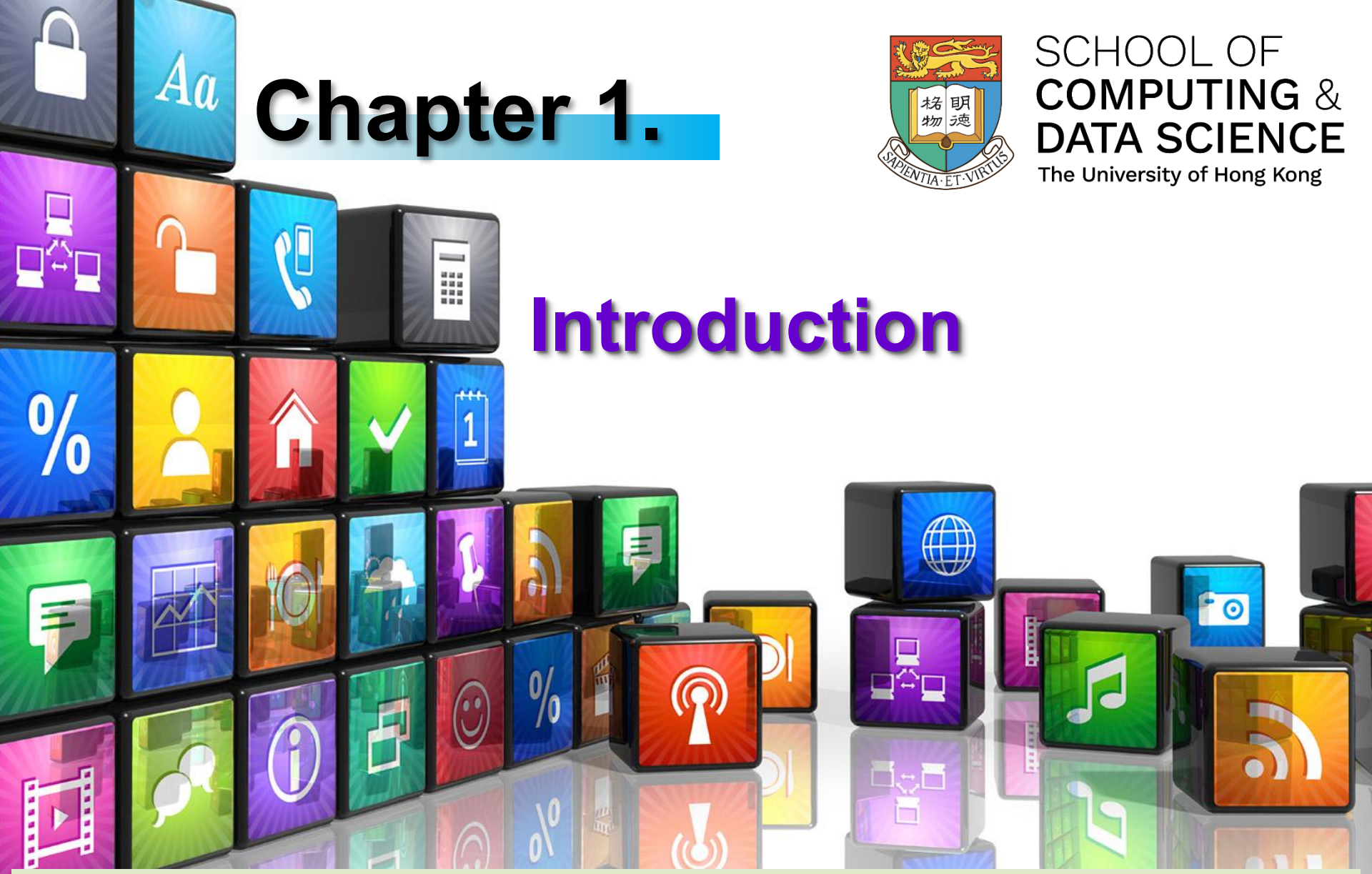
Chapter 1.

SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong

Introduction



2026-2027
COMP3330 Interactive Mobile Application Design and Programming
Dr. T.W. Chim (E-mail: twchim.cs@hku.hk)

Agenda

- Mobile Operating Systems
- Why Mobile Apps?
- Key Principles of Mobile User Experience Design
- UX Constraints of Mobile Apps
- Types of Mobile Apps
- Implication of Mobile Apps to Economy
- Mobile App Development Trends

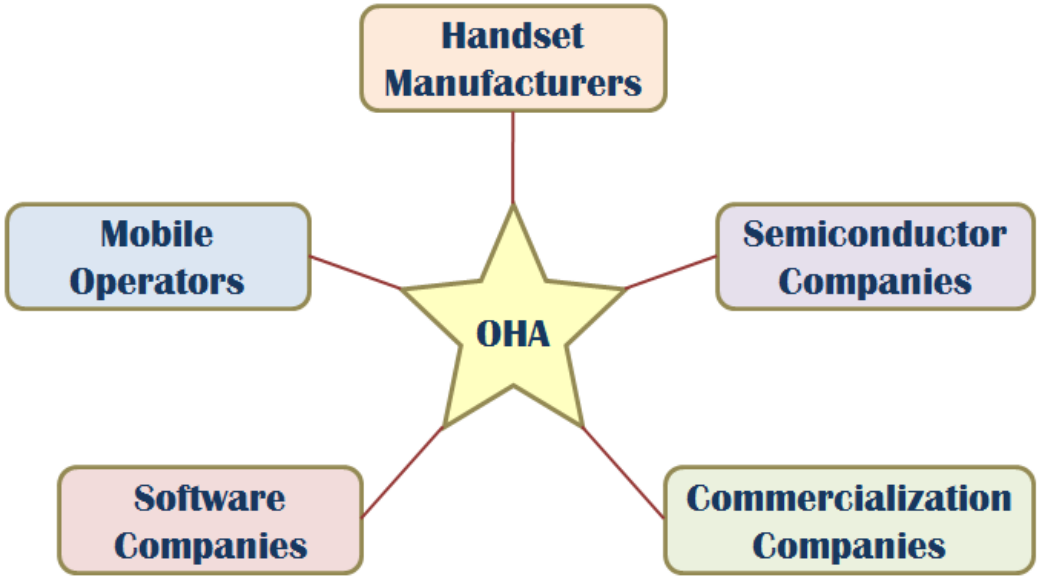
Android



- A software platform and operating system for mobile devices
- Based on the Linux kernel, uses Linux for its device drivers, memory management, process management, and networking
- Developed by the Open Handset Alliance (OHA)

Open Handset Alliance (OHA)
[開放手機聯盟]

- A consortium of over 80 technology and mobile companies, led by Google, established in 2007
- Primary mission: to develop open standards for mobile devices and spearhead the advancement of the Android operating system



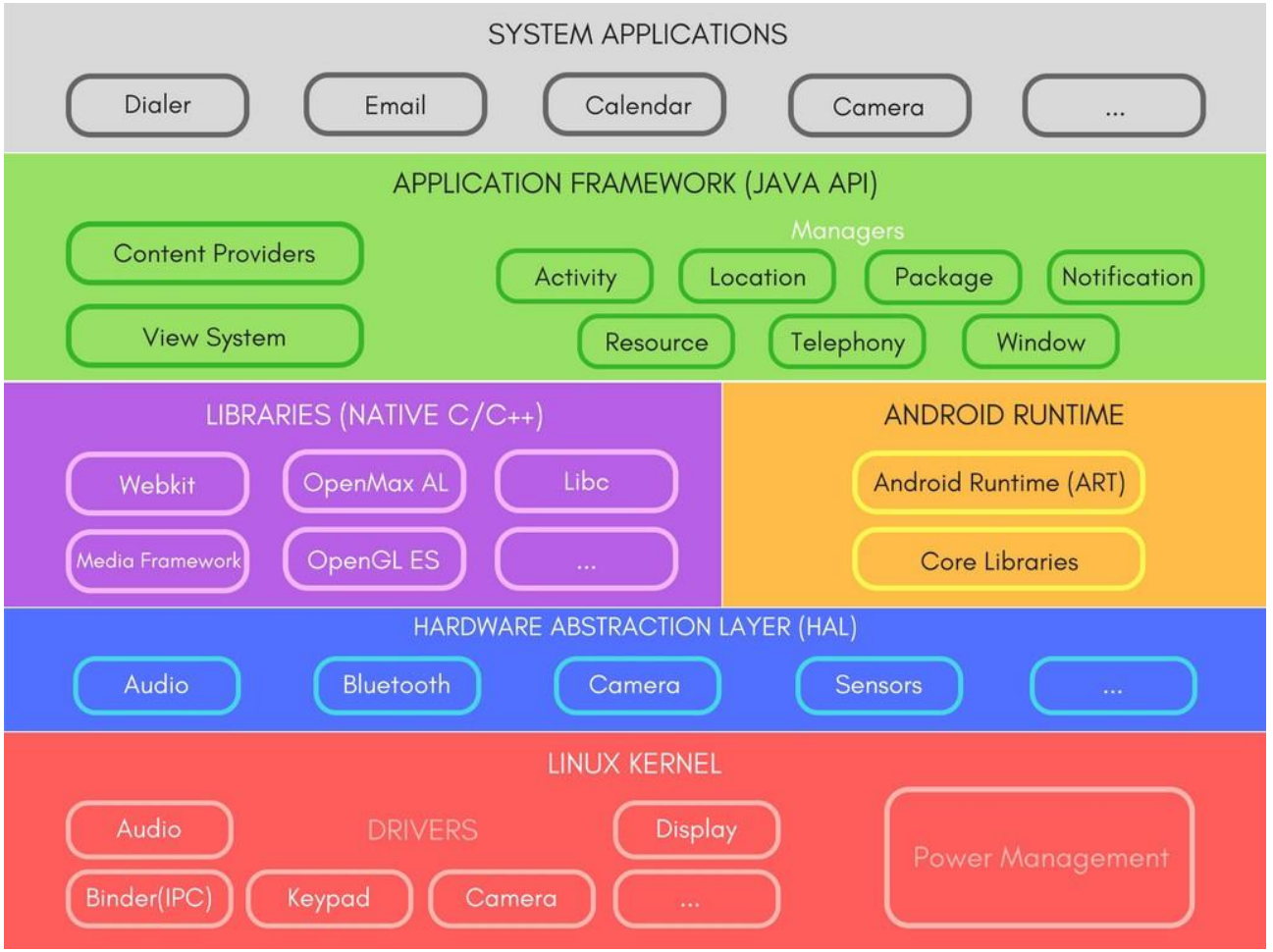
Examples

- Technology Giants:
  - Google, Intel, NVIDIA
- Handset Manufacturers:
  - Samsung, Sony, HTC, Motorola
- Semiconductor Companies:
  - Qualcomm, Texas Instruments
- Mobile Operators:
  - T-Mobile, Sprint

https://www.openhandsetalliance.com/index.html

Architecture





Dalvik Virtual Machine & Android Runtime (ART)

- Managed runtime used by applications and some system services on Android.
- Android 4.4 (KitKat) brought a technology preview of ART, including it as an alternative runtime environment and keeping Dalvik as the default virtual machine. In the subsequent major Android release, Android 5.0 (Lollipop), Dalvik was entirely replaced by ART.
- Key difference:
  - Dalvik: interpretation and trace-based JIT compilation
  - Android Runtime (ART):
    - ahead-of-time (AOT) compilation, compiling entire applications into native machine code upon their installation ➔ improves the overall execution efficiency and reduces power consumption, which results in improved battery autonomy on mobile devices
    - faster execution of applications, improved memory allocation and garbage collection (GC) mechanisms, new applications debugging features, and more accurate high-level profiling of applications

Android



- Security:
  - Android is a multi-process system, in which each application (and parts of the system) runs in its own process. Most security between applications and the system is enforced at the process level through standard Linux facilities, such as user and group IDs that are assigned to applications.
  - Additional finer-grained security features are provided through a "permission" mechanism that enforces restrictions on the specific operations that a particular process can perform, and per-URI permissions for granting ad-hoc access to specific pieces of data.
- Programming languages:
  - Java / Kotlin – officially supported
  - C / C++ – as native library that Gradle can package with APK (using build tool CMake), and being called by Java / Kotlin program via Java Native Interface (JNI)

Android Versions







1.5 CupCake



1.6 Donut



2.0 Eclair



2.2 Froyo



2.3 Gingerbread



3.0 Honeycomb



4.0 IC Sandwich



4.1 Jelly Bean



4.4 KitKat



5.0 Lollipop



6.0 Marshmallo



7.0 Nougat



8.0 Oreo



Android 9 Pie



Android 10 Q



Android 11



Android 12



Android 13



Android 14



Android 15



Android 16



Android 17

HarmonyOS

HarmonyOS

- HarmonyOS (鸿蒙 in Chinese) is a distributed operating system developed by Huawei to run on multiple devices.
- HarmonyOS was first released in September 10, 2019.
- Official website: https://www.openharmony.cn/
- Not Android-based
- ArkCompiler (ARK): proprietary, multi-language compilation and runtime environment, replacing Android's reliance on the Android Runtime (ART) and Java/Kotlin-based architecture. ArkCompiler compiles code directly into native machine instructions to optimize system performance and reduce memory usage.
- HUAWEI AppGallery: official app distribution platform for HUAWEI devices

HarmonyOS

HarmonyOS

- To facilitate application development, several kits are provided:
  - Account Kit
    - Give users easy, quick, and secure sign-ins
  - Location Kit
    - Pinpoint precise user locations with the speed and accuracy of multiple methods
  - Game Service
    - Focus on innovating the gaming experience with in-depth services
  - Scan Kit
    - Give users access to the QR and other barcodes around them
  - Machine Learning Kit
    - Take advantages of machine learning for more visual and language services
  - Awareness Kit
    - Get the context and status of users to give them a smarter experience
  - Push Kit
    - Improve user awareness and engagement in apps
  - Wallet Kit
    - Be part of the All-in-One Wallet digital lifestyle with the convenience of multiple open and accessible passes

Other Android Variations

- One UI (Samsung)
  - Android-based
  - Key features: deep Galaxy ecosystem integration (DeX, SmartThings), polished UX, strong privacy dashboards
- OriginOS (Vivo)
  - Android-based
  - Key features: highly customizable widgets and scenes, gesture-driven navigation, strong visual polish
- ColorOS (OPPO)
  - Android-based
  - Key features: extensive theming / AI features, cross-device tools (O-Share, Multi-Screen Connect), clear privacy controls

Other Android Variations

- HyperOS (Xiaomi)
  - Android-based
  - Key features: streamlined, fast UI with minimal bloat, strong performance / battery focus, cohesive experience with ColorOS lineage
- OxygenOS (OnePlus)
  - Android-based
  - Key features: clean, stock-like UI focused on speed and efficiency, minimal bloat and strong developer-friendly experience.
  - Note: many newer OnePlus devices now use HyperOS
- MagicOS (Honor, sub-brand of Huawei)
  - Android-based
  - Key features: cross-device experience, AI enhancements, smooth UI, cloud integration and multi-device workflows.

iOS (iPhone OS)

- iOS (known as iPhone OS prior to June 2010) is Apple's mobile operating system.
- Apple is a company who developed iPhone Operating System (iOS).
- This OS was released on June 29, 2007, with the first iPhone.
- Used in iPhone, iPad, iPod touch, Apple Watch and Apple TV.



First iPhone on display under glass at 2007 Macworld

iOS (iPhone OS)

- Game center:
  - A service introduced by Apple that allows users to play and challenge friends when playing online multiplayer social gaming network games. Games can now share multiplayer functionality between the Mac and iOS versions of the app.
  - Game Center was announced during an iOS 4 preview event hosted by Apple on April 8, 2010. A preview was released to registered Apple developers in August 2010.

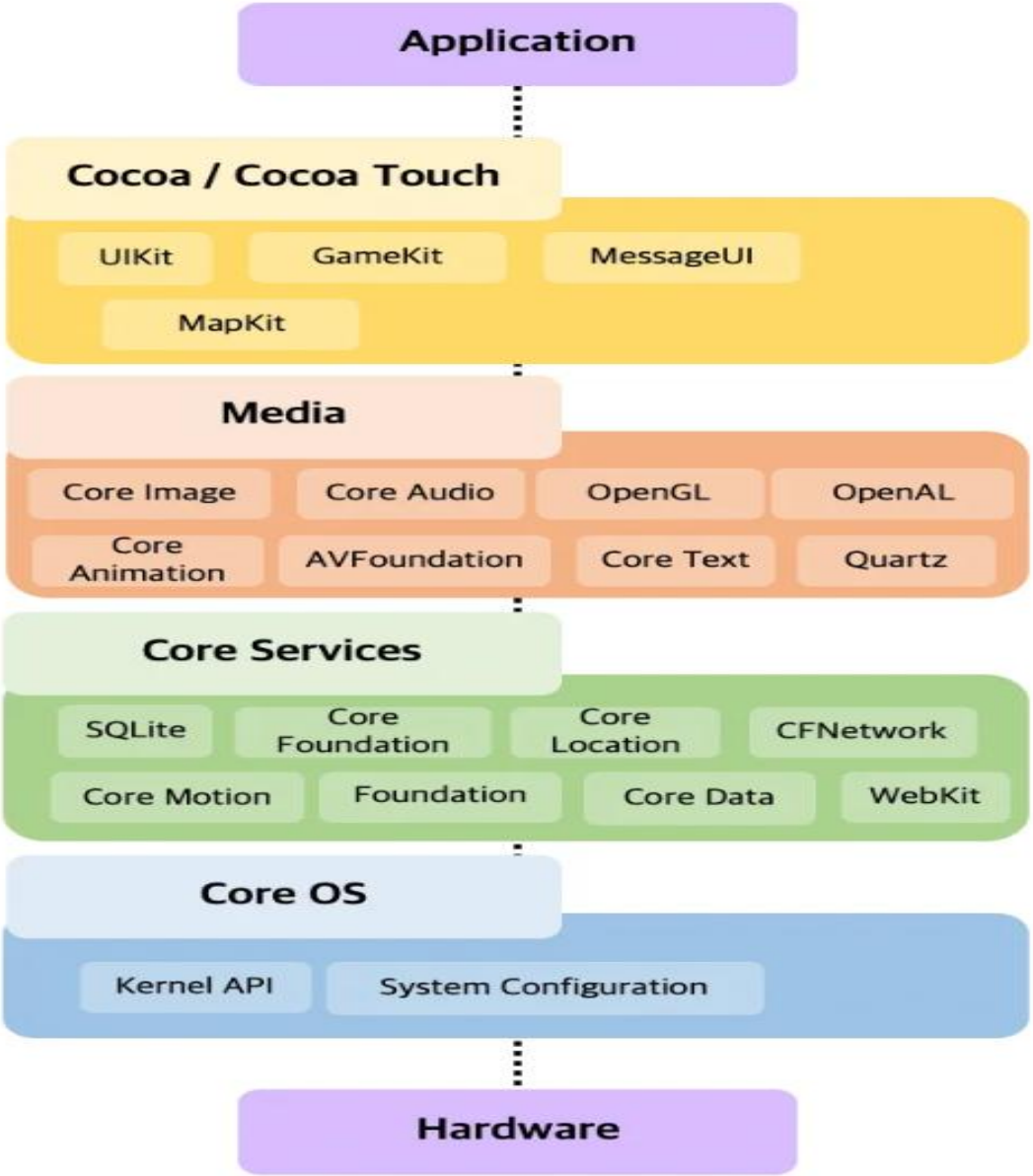
- Programming languages:
  - Objective-C – most common
  - Swift – the emerging programming language
  - C / C++ – also possible but not supported

Sample functions:

- **Leaderboards**
- **Dashboard**
- **Achievements**
- **Challenges**
- **Multiplayer**
- **Friends**



Technology Layers



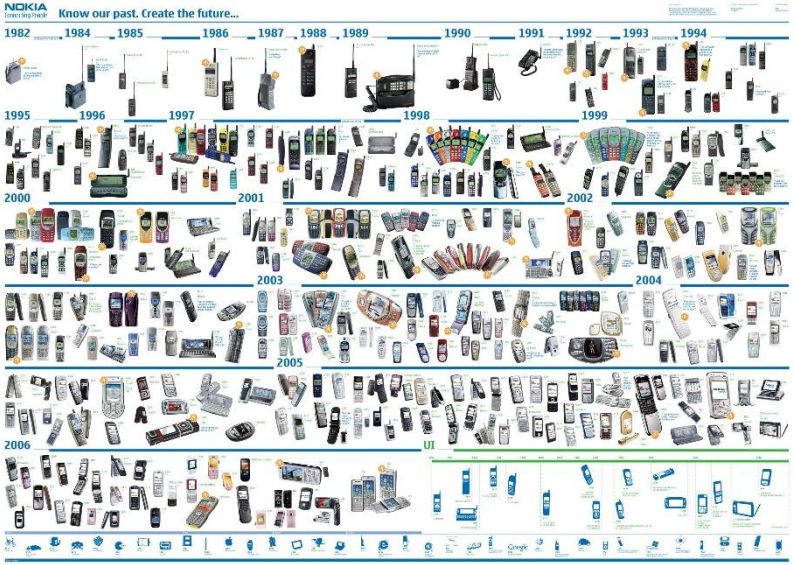
BlackBerry OS

- BlackBerry OS is a proprietary mobile operating system developed by the Canadian company BlackBerry Limited (formerly known as Research In Motion, or RIM) in January 1999.
- BlackBerry provides enterprises and governments with the software and services they need to secure the Internet of Things (e.g., e-mail service).
- Physical keyboard is a feature of BlackBerry devices.
- In 2013, BlackBerry introduced BlackBerry 10, a major revamp of the platform based on the QNX operating system. BlackBerry 10 was meant to replace the aging BlackBerry OS platform with a new system that was more in line with the user experiences of Android and iOS platforms.
- In September 2015, BlackBerry announced the Priv, which is an android-based handset.



Symbian OS

- Produced by the software development and licensing company Symbian Ltd.
- Symbian Ltd. was established in June 1998 and is headquartered in Southwark in the UK.
- The native language of the Symbian OS is C++.
- Java 2nd the most important programming language on Symbian OS.
- Mostly used in Nokia's mobile phones
- The last Nokia phone supporting Symbian was Nokia 808 PureView, which was released in 2012.
- Final release: Nokia Belle Feature Pack 2 (released on 2 Oct, 2012)





Windows Mobile OS / Windows Phone OS

- Windows Mobile (2000 – 2013):
  - A mobile operating system developed by Microsoft
  - Last version: Windows Mobile 6.5 (based on Windows CE 5.2 kernel)
  - Superseded by Windows Phone
- Windows Phone (2010 – 2015):
  - Latest version: Windows 10 Mobile
- Feature: Coherent file system similar to that of Windows 9x/Windows NT and support many of the same file types.
- Note: Windows is providing more support to Android and iOS than before.
  - Windows 10 users can use Microsoft's Your Phone app to mirror phone content to a PC (https://www.theverge.com/2018/8/14/17689670/microsoft-windows-10-your-phone-app-microsoft-store-download)



KaiOS

- KaiOS is a lightweight mobile operating system designed for keypad-based mobile phones, combining basic phone simplicity with smartphone-like capabilities.
- Initial release: October 2017
- Latest version: 4.0 (2025), Android 14 base
- Key Features
  - Supports 4G LTE, 5G, Wi-Fi, GPS, and VoLTE
  - Runs popular apps like YouTube, Google Maps, and Facebook
  - Optimized for low-cost hardware
  - Long battery life and simple keypad-based navigation
  - App ecosystem powered by HTML5-based apps
- Example Devices
  - JioPhone
  - Nokia 8110 4G
  - Alcatel Go Flip series





Market Share of Mobile OS (August 2026)

Worldwide					
Android	iOS	Linux	KaiOS	Windows	Samsung
67.61%	32.36%	0.01%	0.01%	0.01%	0%
China					
Android	iOS	Windows	Blackberry	SymbianOS	Linux
71.34%	28.58%	0.06%	0.02%	0.01%	0%
United States of America					
iOS	Android	KaiOS	Windows	Linux	Samsung
60.68%	39.29%	0.02%	0%	0%	0%

https://gs.statcounter.com/os-market-share/mobile/worldwide
https://gs.statcounter.com/os-market-share/mobile/china
https://gs.statcounter.com/os-market-share/mobile/united-states-of-america

Market Share of Programming Languages

Worldwide, Aug 2026 :

Rank	Change	Language	Share	1-year trend
1		Python	50.44 %	+20.2 %
2		Java	12.8 %	-2.0 %
3		C/C++	8.57 %	-1.2 %
4	↑	R	4.18 %	-1.3 %
5	↓	JavaScript	3.52 %	-3.4 %
6	↑↑↑↑↑↑	Objective-C	3.01 %	+0.9 %
7		PHP	2.06 %	-1.4 %
8	↑	Rust	1.86 %	-1.0 %
9	↓↓↓	C#	1.74 %	-3.0 %
10	↓↓	Swift	1.68 %	-1.2 %
11	↓	TypeScript	1.52 %	-0.9 %
12	↓	Ada	1.5 %	-0.7 %
13		Matlab	0.99 %	-0.5 %
14	↑↑	Powershell	0.8 %	-0.3 %
15	↑↑	Ruby	0.77 %	-0.3 %
16	↑↑	VBA	0.66 %	-0.3 %
17	↑↑↑	Dart	0.55 %	-0.4 %
18	↓↓↓	Kotlin	0.54 %	-0.9 %
19		Lua	0.52 %	-0.4 %
20	↓↓↓↓↓↓	Go	0.49 %	-1.0 %
21	↑	Julia	0.46 %	-0.0 %
22	↑	Scala	0.23 %	-0.2 %
23	↑↑↑↑	Delphi/Pascal	0.22 %	+0.1 %
24	↓↓↓	Abap	0.18 %	-0.3 %
25	↑	Zig	0.15 %	+0.0 %
26	↓↓	Visual Basic	0.15 %	-0.2 %
27	↑↑	Perl	0.14 %	+0.0 %
28	↑↑	Haskell	0.12 %	+0.1 %
29	↓	Groovy	0.09 %	-0.0 %
30	↓↓↓↓↓	Cobol	0.05 %	-0.1 %

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Reference: http://pypl.github.io/PYPL.html



Why Mobile Apps?

Why Mobile Apps?

- Comparing with reading a related page using browsers, using smart phone apps have the following advantages:
  - More convenient
    - No need to type in URL especially for the first time
  - Better fit to the mobile phone screen
    - Web pages are designed to be browsed using computers in general. So the words are usually too small to be displayed on mobile phone screens.

Why Mobile Apps?

- Less data is downloaded from the server
  - For browsers, all components including text and images are downloaded from the server.
  - For apps, images are stored in the apps and only essential elements (basically text) are downloaded from the server.
  - For example:
    - 2.9 KB for transmitting the sunny icon
    - 5 B for transmitting the word “sunny”
    - Can further save data: only 39 possible icons and so each icon can be uniquely represented by 6 bits!



10 Reasons Why Apps Are Better

An article published in September 2024:
https://vwo.com/blog/10-reasons-mobile-apps-are-better/

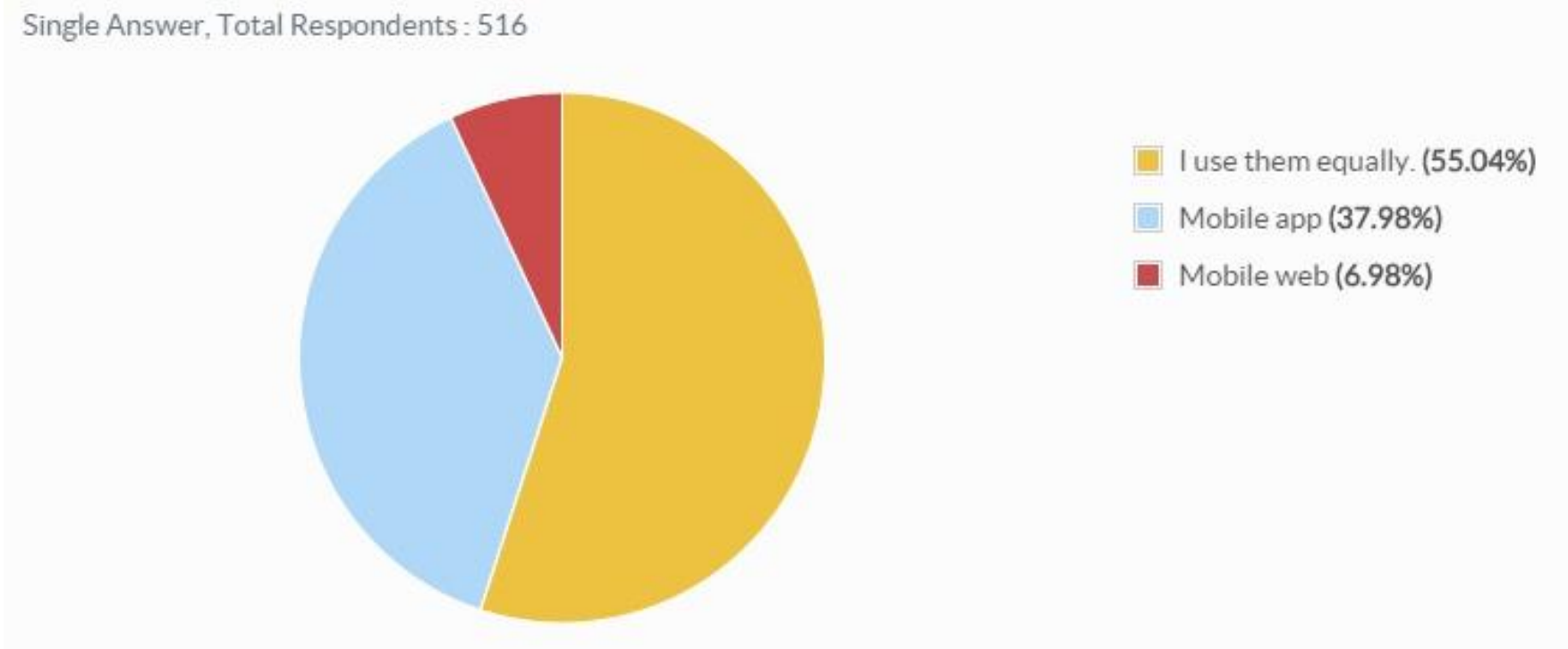
- Mobile apps offer better personalization
  - E.g., based on interests, location, usage behavior
- Ease of sending notifications
  - 2 types: push and in-app notifications
- Making use of mobile device features
  - E.g., camera, contact list, GPS, phone calls, accelerometer, compass
- Ability to work offline
- Freedom in designing
  - E.g., can base on advanced gestures like tap, swipe, drag, pinch, hold

10 Reasons Why Apps Are Better

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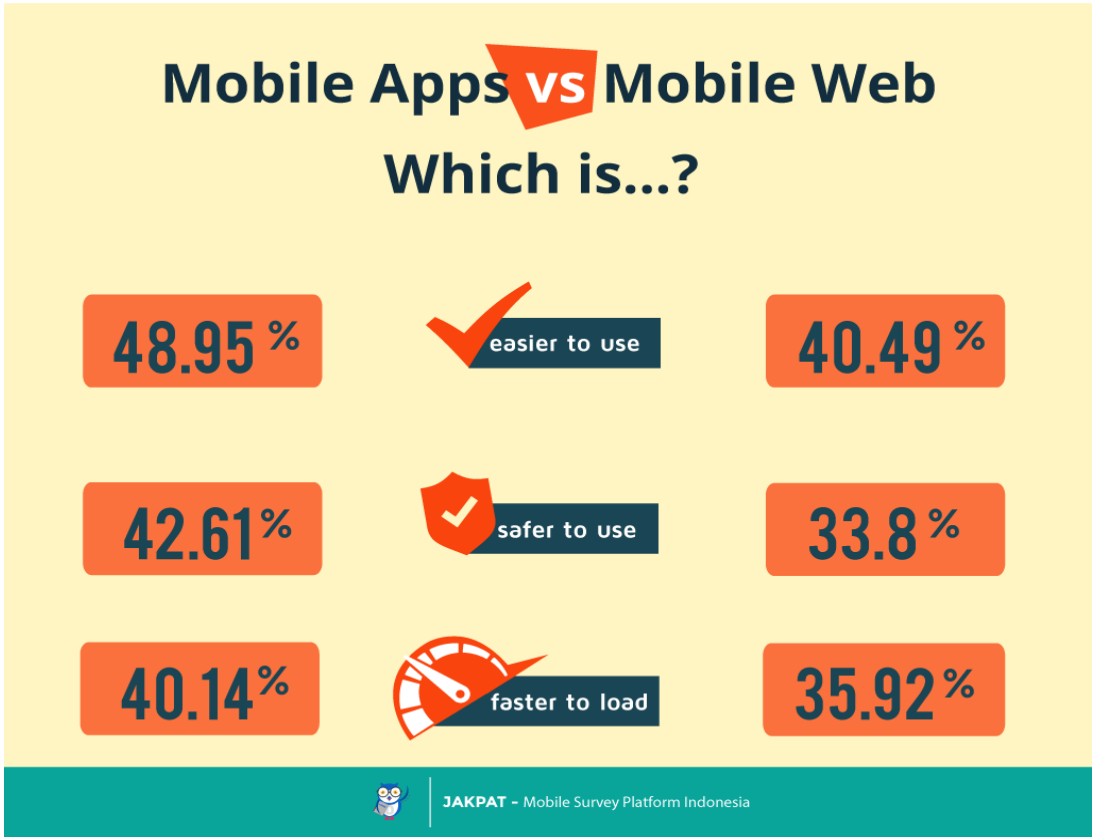
- New branding experience
  - New branding styles to customers
- Brand presence
  - App icon is like a mini-advertisement for the brand
- New stream of conversions
  - Conversion funnel: a roadmap that you create for potential customers.
- Users spend more time on Apps
  - Mobile users spend 88% of their time on mobile apps and just 12% of their time on mobile websites
- Apps can work faster than websites

User's preference in the mobile space



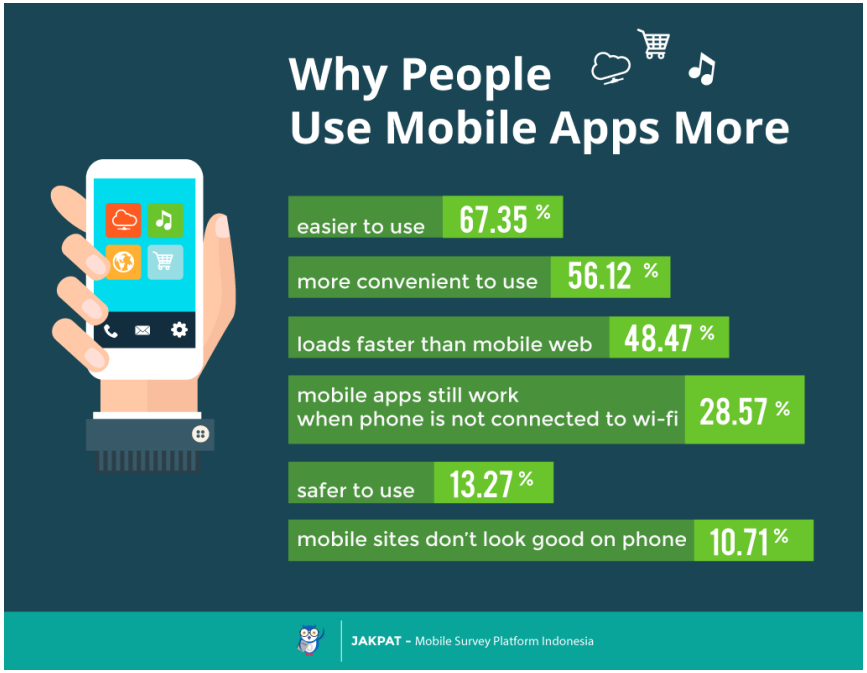
Reference: http://blog.jakpat.net/mobile-web-vs-mobile-apps-survey-report/

User's preference in the mobile space



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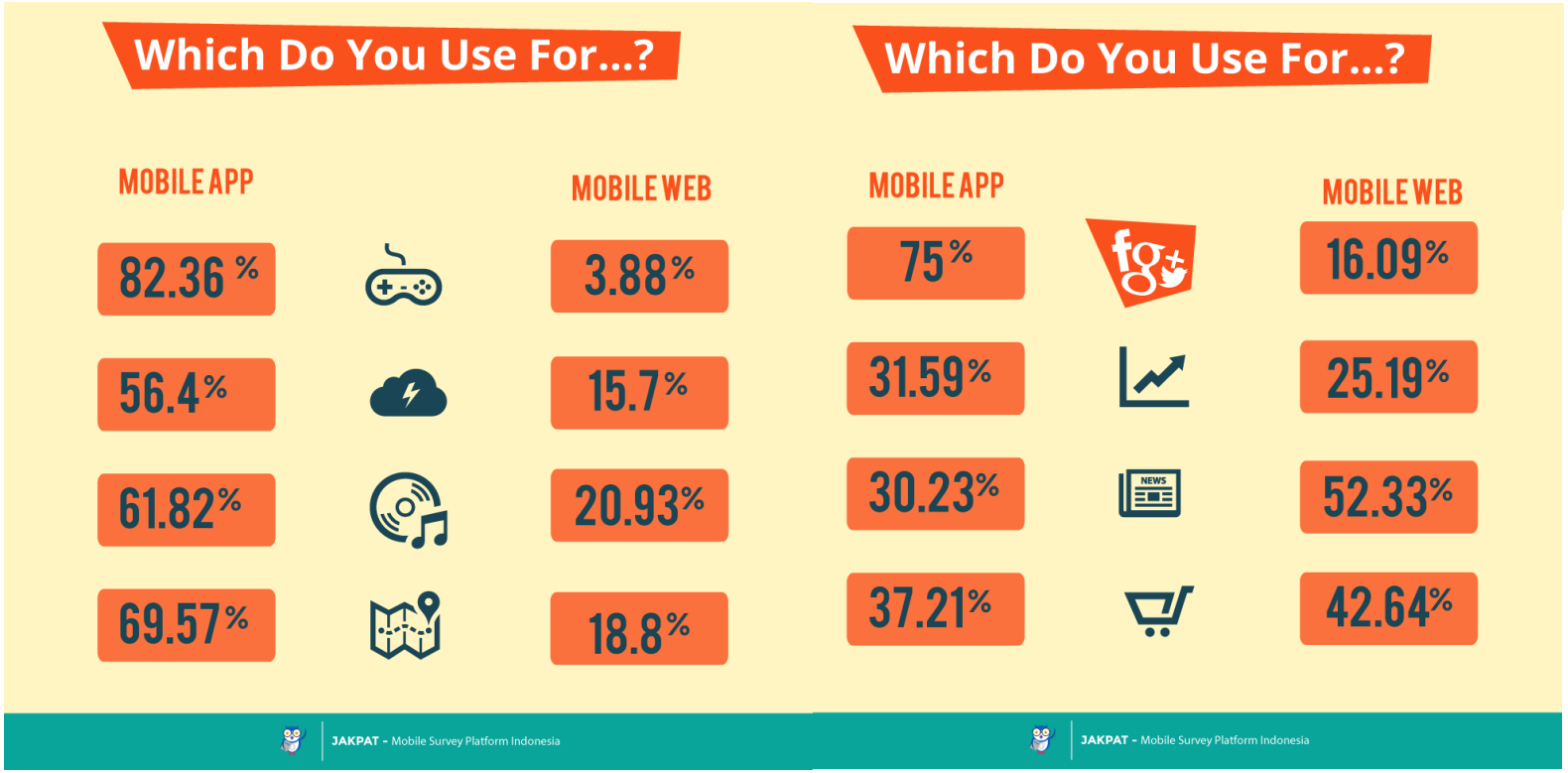
User's preference in the mobile space





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User's preference in the mobile space



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Key Principles of Mobile User Experience Design

Key Principles of Mobile User Experience Design

- Principle 1: Remove the clutter, protect your design from overloads
  - Allocate user's attention accordingly to your or user's needs. Too much information can scare your user away and cause confusion.
  - Get rid of all odd elements such as buttons, images, texts and other elements that are not vital. Perfection is achieved when there is nothing left to take away. Improve the comprehension by providing one primary action per screen. On mobile, it's better to have many screens with a single action per each rather than one screen with many actions that are not clear.
  - 'One of the basic rules of good UX is to reduce the effort users have to put in to get what they want.' – Nick Babich.

Key Principles of Mobile User Experience Design

- Principle 2: Intuitive navigation
  - This principle is at a high priority for every mobile (and web as well) app design. Good navigation should be invisible and effortlessly lead the user to the proper destination without any explanation. Visual metaphors and navigation elements must create a coherent flow.
  - Mobile navigation must be consistent and communicate the current location. The user should know where exactly he or she is right now.

Key Principles of Mobile User Experience Design

- Principle 3: Seamless experience for all type of devices
  - Creating a seamless experience across mobile, desktop and tablet is very important for your users. There should be no dramatic difference for the same app across different devices.
- Principle 4: Design finger-friendly tap-targets
  - All the targets in mobile interfaces design should be big enough so users could tap them easily with a finger. Not less than 7 – 10 mm. Don’t forget about the clear visual feedback.

Key Principles of Mobile User Experience Design

- Principle 5: Legible content
  - One of the challenges of mobile design is to fit a lot of information on a small user interface. Do not try to squish everything down in an attempt to provide as much information as possible. Remember, that text should be at least 11 points to be legible at a typical viewing distance without zooming. A good UI has a lot of breathing room.
  - It's better not to experiment with decorative fonts, but use the standard Helvetica Neue for iOS and Roboto for Android.

Key Principles of Mobile User Experience Design

- Principle 6: Clearly visible interface elements
  - Use color and contrast to help users see and interpret your content. The contrast between elements is especially vital for users with low vision.
  - The W3C recommends the following contrast ratios for body text and image text:
    - Small text should have a contrast ratio of at least 4.5:1 against its background.
    - Large text (at 14 pt bold / 18 pt regular and up) should have a contrast ratio of at least 3:1 against its background.
    - Contrast ratio = (L1 + 0.05) / (L2 + 0.05) where
      - L1 is the relative luminance of the lighter of the colors
      - L2 is the relative luminance of the darker of the colors
      - Relative luminance can be formed as a properly weighted sum of RGB
    - Reference: https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html#dfn-image-of-text

Key Principles of Mobile User Experience Design

- Principle 7: Design controls should be based on hand position
  - According to Steven Hoober research, about 49% of mobile users rely on one thumb to get things done on their phones. So when designing a mobile app, take care about comfort zones for a person's one-handed reach on a smartphone. Place all the top-level menus, frequently-used controls and common actions in these comfort zones of one-thumb interactions.

Key Principles of Mobile User Experience Design

- Principle 8: Minimize the need for typing
  - Keep forms as short and simple as possible by removing any unnecessary fields. Use auto-complete and personalized data where appropriate so that users only have to enter the bare minimum of information.



UX Constraints of Mobile Apps

UX Constraints of Mobile Apps

- Constraint 1: Client-side storage
  - One of the major constraints of mobile apps is that the app data sits client side rather than server side like on web (think Amazon, Facebook, Twitter, etc.). That means that every time a user downloads an app, they lose a certain amount of storage.
  - When users use up the storage, users often have to make a quick decision. They have to delete something. So the question becomes, is your app more important to your users than a family video, or favorite song?
  - On mobile, frequency of use is far more important than perceived utility…

UX Constraints of Mobile Apps

- Constraint 2: Small screen sizes and clunky controls
  - Even as screen sizes become larger and larger, the overall mobile experience is still not an optimal method for reading or accomplishing tasks.
  - “Mobile screens are smaller: reading through a peephole increases cognitive load and makes it about twice as hard to understand.” - Jakob Nielsen, Web Utility Consultant.
  - In addition, the controls and keyboards on smartphones can be particularly difficult to use. According to Foolproof UX, 56% of users said they had not signed up for an app or mobile service because the registration process was too time consuming.

UX Constraints of Mobile Apps

- Constraint 3: Environments full of distractions
  - Mobile users are much more prone to dropping off than their web counterparts.
    - Personal computers are more commonly used in workspaces such as coffee shops, home offices, and at work.
    - Phones are often used in non-work settings.
  - Even if they intend to return later, users often forget and never come back, deleting the app after weeks without use.
  - Once users drop off, you need to have a strategy for bringing them back into your app…

Mobile v.s. Web





UX Constraints of Mobile Apps

- Constraint 4: Making any update is incredibly hard
  - “Shipping mobile software is inherently different than shipping web software – the stakes are higher.” - Christian Legnitto, Release Manager at Facebook
  - Making even the tiniest of changes in a mobile app is incredibly hard. Most teams already know that shipping even a small change takes an average of 7 days to be approved by Apple. Furthermore, because the data sits client side, users have to go in and download those updates manually.
  - If changes don’t perform as expected (driving wrong user behaviors, introducing bugs, etc.), fixing the issue immediately is out of the question. Instead, you’ll have to wait for an additional review process and slow user adoption.
  - That’s why release management is so important.

UX Constraints of Mobile Apps

- Constraints can be catalysts
  - On mobile, there are a lot of constraints to overcome, but as we've seen, constraints can actually make you more creative.
  - While on mobile you'll have to deal with small screen sizes, limited storage and connectivity, distractions, and app stores they'll also force your team to create better products.



Types of Mobile Apps

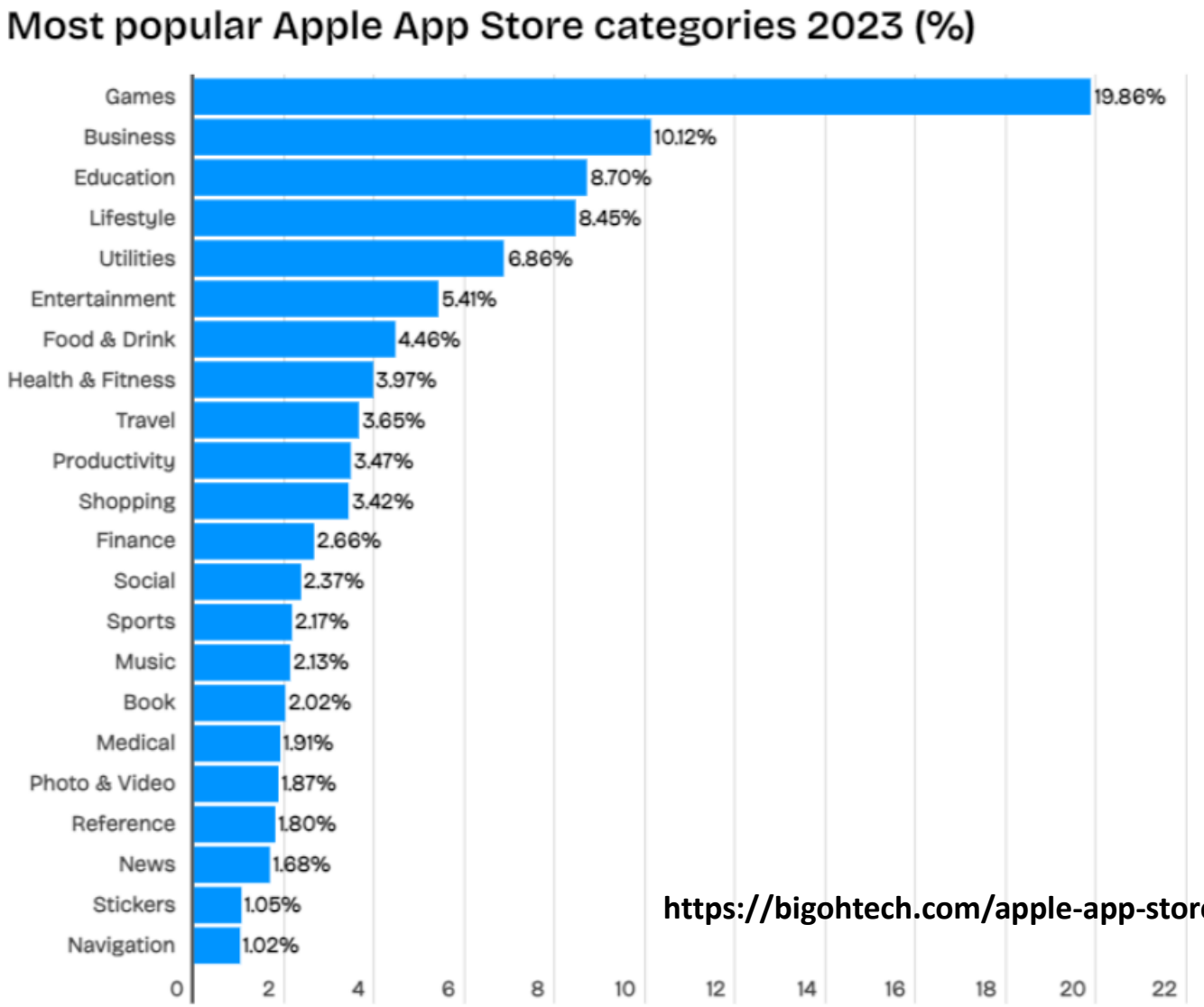
By Categories

- Game
- Weather
- Social Networking
- Navigation / Map
- Music
- News
- Entertainment
- Sports
- Banking / Finance
- Shopping / Retail
- Dining / Restaurant
- Search
- Search and Bid
- Digital Books / Magazines
- …



Google Play: https://play.google.com/store/apps
Apple Store: http://www.apple.com/hk/itunes/

Most popular Apple App Store categories in 2023, by share of available apps



https://bigohtech.com/apple-app-store-statistics/

By Nature

- Native Apps
  - Native apps are what typically springs to mind when you think of an app.
  - You download them from the App Store or Google Play, they sit within your device's applications, and you launch them by tapping their icon.
- Web Apps
  - Web app is the mobile version of a site.
  - An "app" like this loads within a mobile browser, like Safari or Chrome, like every other website. Your audience doesn't have to install a web app. They don't need to have available space on their devices.
  - Web apps are sometimes designed to look and behave like apps and are in general ideal when the purpose is simply to make content or functionality available on mobile, but an app is either not a good fit or too expensive.
  - Example: http://www.cs.hku.hk/

By Nature

- Hybrid Apps
  - The bulk of the app is built using cross-compatible web technologies, such as HTML5, CSS and JavaScript — the same languages used to write web apps.
  - Some native code is used however to allow the app to access the wider functionality of the device and produce a more refined user experience. For native apps, instead only native code is used.
  - The advantage of this approach is obvious: only a portion of native code has to be re-written to make the app work on the different kinds of devices available.
  - They wrap “web views” within native code to deliver their content.

```
<WebView  xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/webview"
   android:layout_width="fill_parent"
   android:layout_height="fill_parent"
/>
```

```
WebView browser = (WebView) findViewById(R.id.webview);
```

```
browser.loadUrl("http://www.tutorialspoint.com");
```

By Nature

NATIVE vs. WEB vs. HYBRID: 7 FACTORS OF COMPARISON

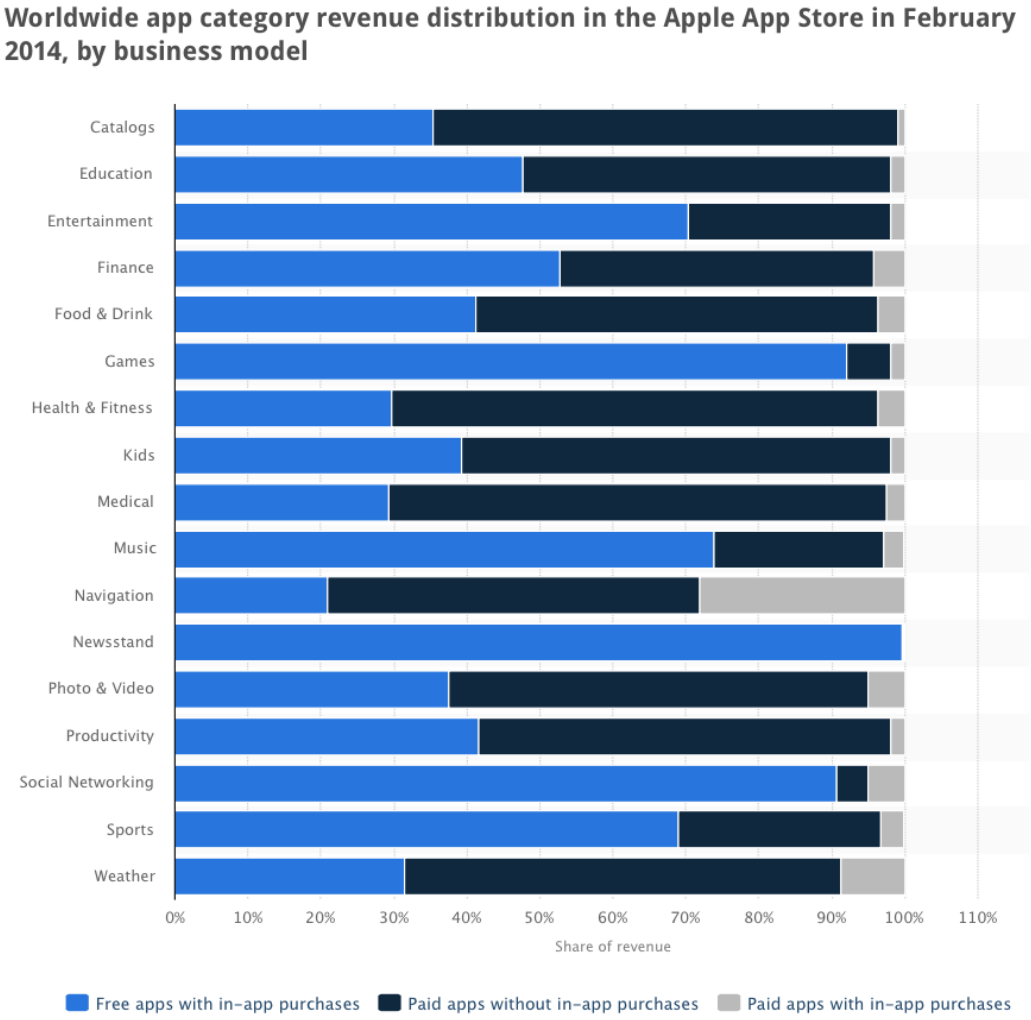
KEY: CON | PRO | NEUTRAL

	NATIVE	HYBRID	WEB
COST	Commonly the highest of the three choices if developing for multiple platforms	Similar to pure web costs, but extra skills are required for hybrid tools	Lowest cost due to single codebase and common skillset
CODE REUSABILITY/ PORTABILITY	Code for one platform only works for that platform	Most hybrid tools will enable portability of a single codebase to the major mobile platforms	Browser compatibility and performance are the only concerns
DEVICE ACCESS	Platform SDK enables access to all device APIs	Many device APIs closed to web apps can be accessed, depending on the tool	Only a few device APIs like geolocation can be accessed, but the number is growing
UI CONSISTENCY	Platform comes with familiar, original UI components	UI frameworks can achieve a fairly native look	UI frameworks can achieve a fairly native look
DISTRIBUTION	App stores provide marketing benefits, but also have requirements and restrictions	App stores provide marketing benefits, but also have requirements and restrictions	No restrictions to launch, but there are no app store benefits
PERFORMANCE	Native code has direct access to platform functionality, resulting in better performance	For complex apps, the abstraction layers often prevent native-like performance	Performance is based on browser and network connection
MONETIZATION	More monetization opportunities, but stores take a percentage	More monetization opportunities, but stores take a percentage	No store commissions or setup costs, but there are few monetization methods

Source: https://static.dzone.com/dz1/dz-files/Screen%20Shot%202014-06-13%20at%205.05.49%20PM.png

By Business Model

- Free apps with in-app purchases
- Paid apps without in-app purchases
- Paid apps with in-app purchases



Reference: http://www.statista.com/statistics/283753/apple-app-store-global-category-revenue-share-per-business-model/



Mobile App Development Trends

Sharing of Economy

- An activity of acquiring, providing, or sharing access to goods and services that is often facilitated by a community-based online platform.
- Sharing economy has succeeded in large part because the real economy has been struggling
  - Lots of people are trying to fill holes in their income by monetizing their stuff and their labor in creative ways [Source: http://nymag.com/daily/intelligencer/2014/04/sharing-economy-is-about-desperation.html]
  - Mobile and e-payment technology helps boosting sharing economy
- The users of a Sharing Economy platform have to trust the platform itself as well as the product at hand

App Store > Travel > Airbnb, Inc.

Airbnb 4+

Airbnb, Inc. >

Essentials

Offers iMessage App for iOS Offers Apple Watch App for iPhone

Details Ratings and Reviews Related

Download

This app is designed for both iPhone and iPad

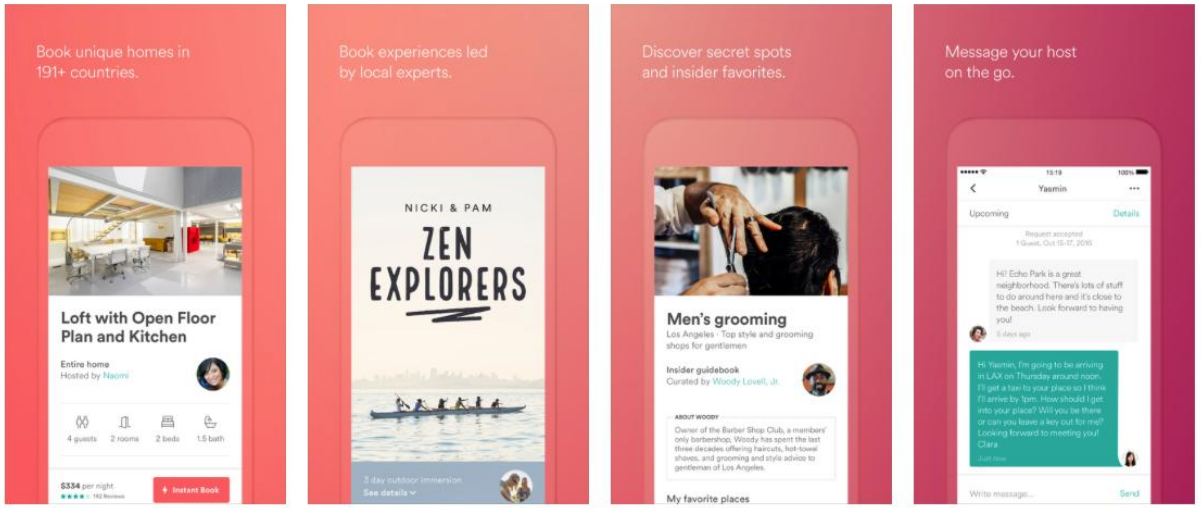
Rating: 4+

LINKS

Privacy Policy
Developer Website

©2015 Airbnb, Inc.

Screenshots iPhone iPad iMessage Apple Watch Apple TV





- Open up homes to complete strangers for money
- Allows everyone to run their mini-hotel.
- Airbnb is an online marketplace and hospitality service, enabling people to list or rent short-term lodging including vacation rentals, apartment rentals, homestays, hostel beds, or hotel rooms.
- The company does not own any lodging; it is merely a broker and receives percentage service fees from both guests and hosts in conjunction with every booking.
- It has over 3,000,000 lodging listings in 65,000 cities and 191 countries, and the cost of lodging is set by the host.

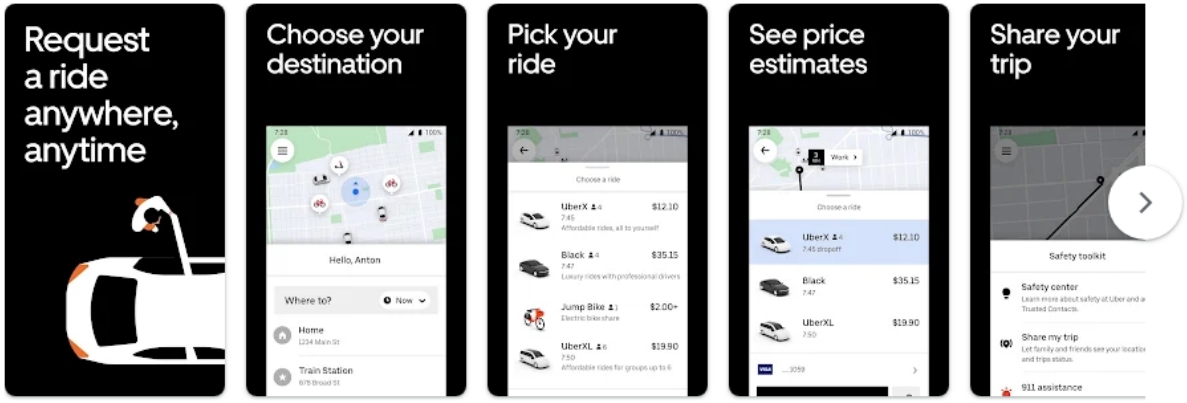
Uber - Request a ride

Uber Technologies, Inc.

4.3★ 10.2M reviews | 500M+ Downloads | 3+ Rated for 3+

Install | Add to wishlist







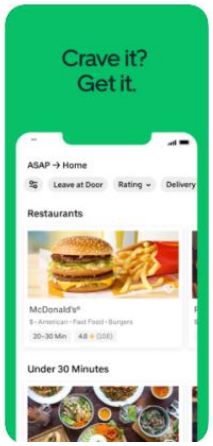
- Open up cars to strangers for money
- Allows everyone to be drivers
- Uber Technologies Inc. is an American worldwide online transportation network company headquartered in San Francisco, California.
- It develops, markets and operates the Uber app, which allows consumers with smart phones to submit a trip request, which the software program then automatically sends to the Uber driver nearest to the consumer, alerting the driver to the location of the customer.
- Uber drivers use their own personal cars.

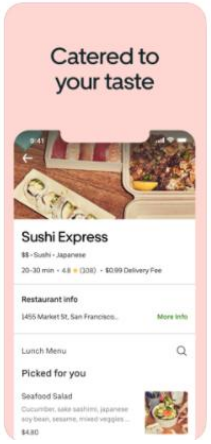


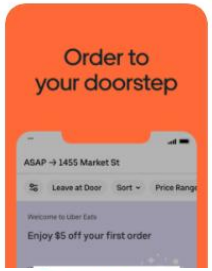
- The Uber app automatically calculates the fare and transfers the payment to the driver.
- In 2015, Uber achieved its 1 billion rides milestone since its founding
- As of August 2016, the service was available in over 66 countries and 545 cities worldwide.
- Since Uber's launch, several other companies have replicated its business model, a trend that has come to be referred to as "Uberification".
- The legality of Uber has been challenged by governments and taxi companies, who allege that its use of drivers who are not licensed to drive taxicabs is unsafe and illegal.
- Uber introduced Uber Taxi in 2021.

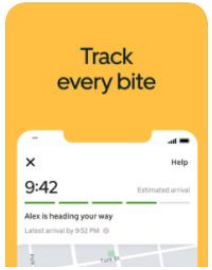
Uber Eats: Food Delivery 4+
Local restaurants to your door
Uber Technologies, Inc.
#2 in Food & Drink
4.8, 1.6M Ratings
Free

Screenshots iPhone iPad

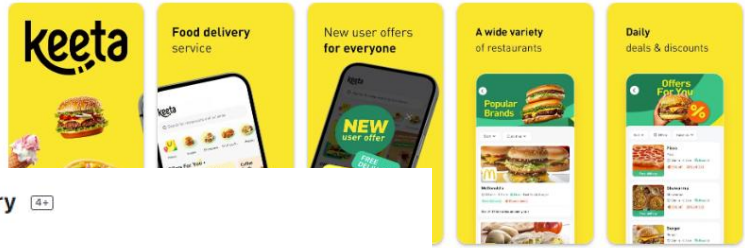








Keeta - Food Delivery
KANGAROO LIMITED
4.6★ 90.9K reviews
1M+ Downloads
3+ Rated for 3+
This app is available for some of your devices



foodpanda - Food Delivery 4+
foodpanda
Foodpanda
#153 in Food & Drink
4.6, 92.5K Ratings
Free

iPhone Screenshots





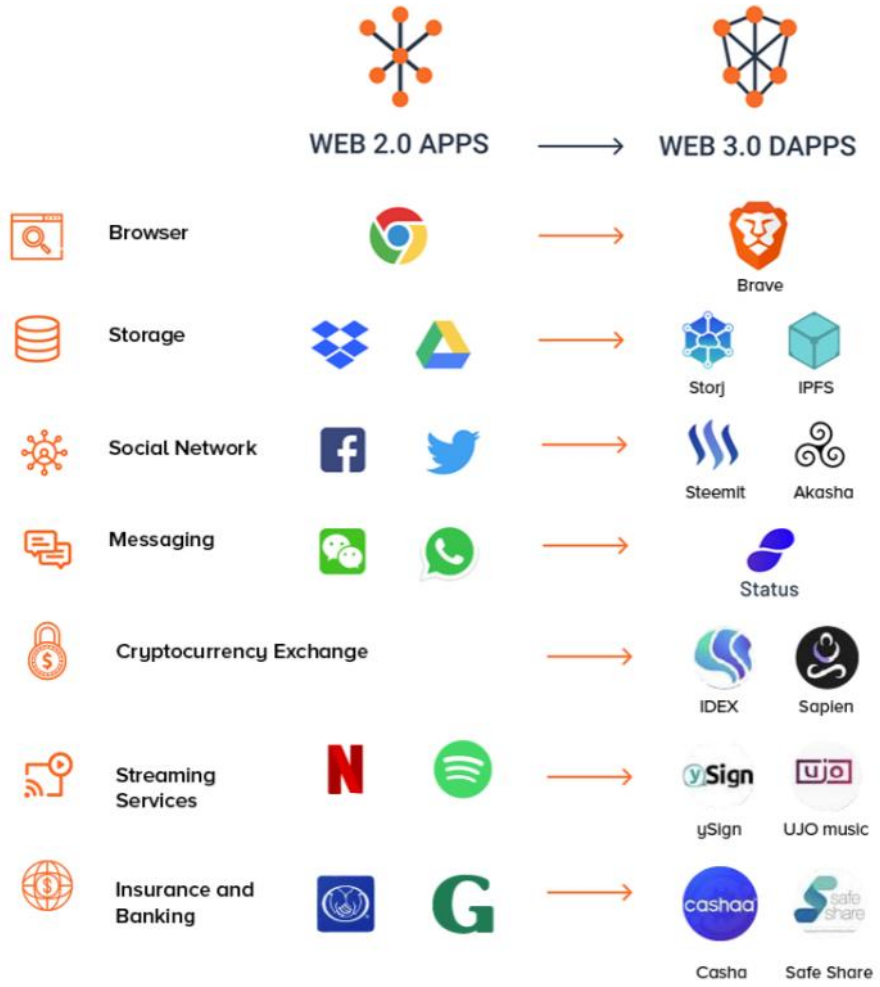




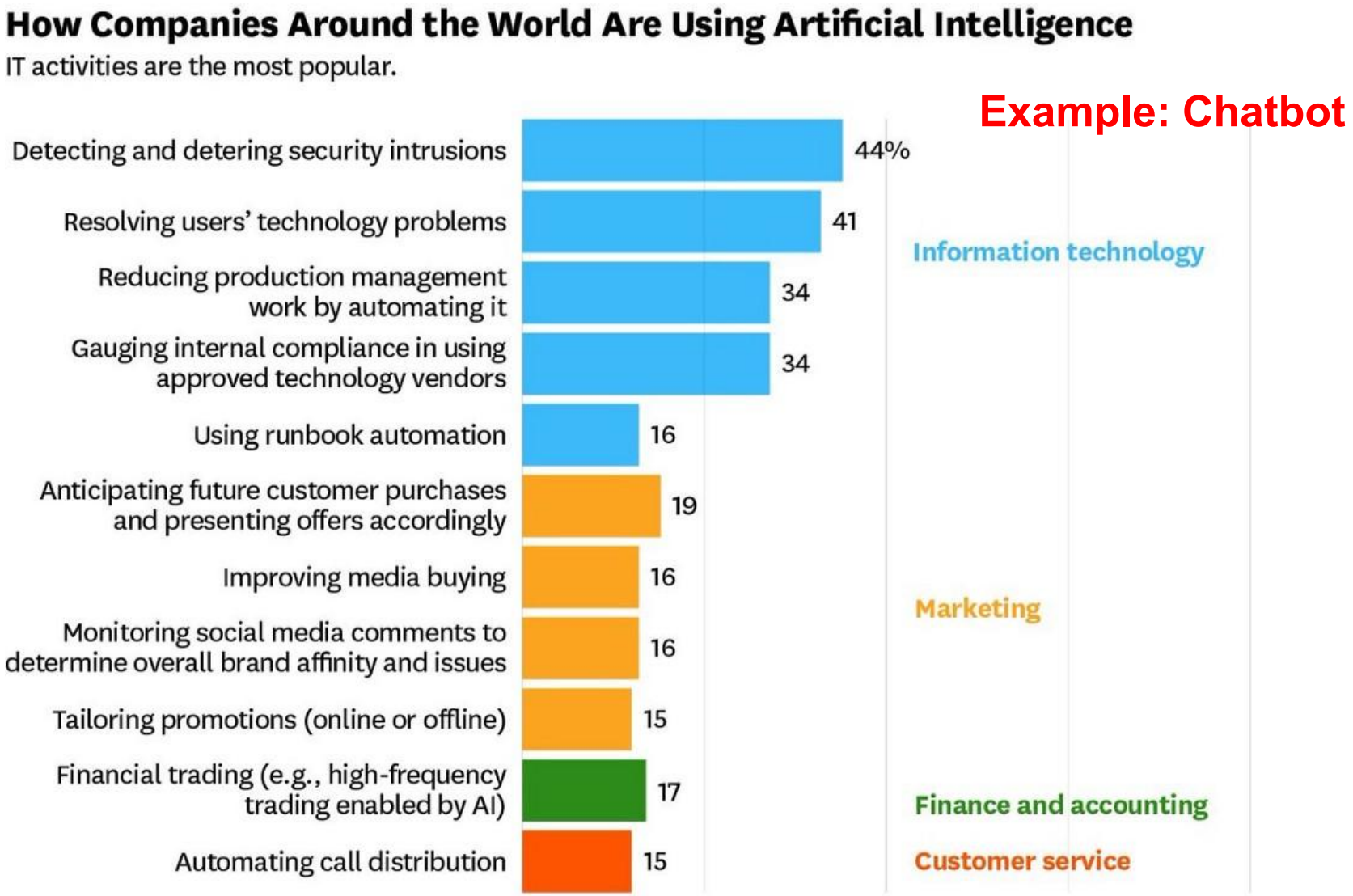
- Allows small restaurants to expand their reach to a larger customer base.
- On-demand meal delivery service.
- Key Components:
  - **Restaurant Partners:** Teams up with restaurants in different areas to offer a range of cuisines
  - **Menu:** May have daily rotating menu
  - **Ordering:** Customers can order through apps
  - **Payment:** Payment can be cashless, credit card, etc.
  - **Delivery:** Meals are delivered to the location selected in the app by couriers — who may be in cars, on foot, or on bikes.

Web3 Applications

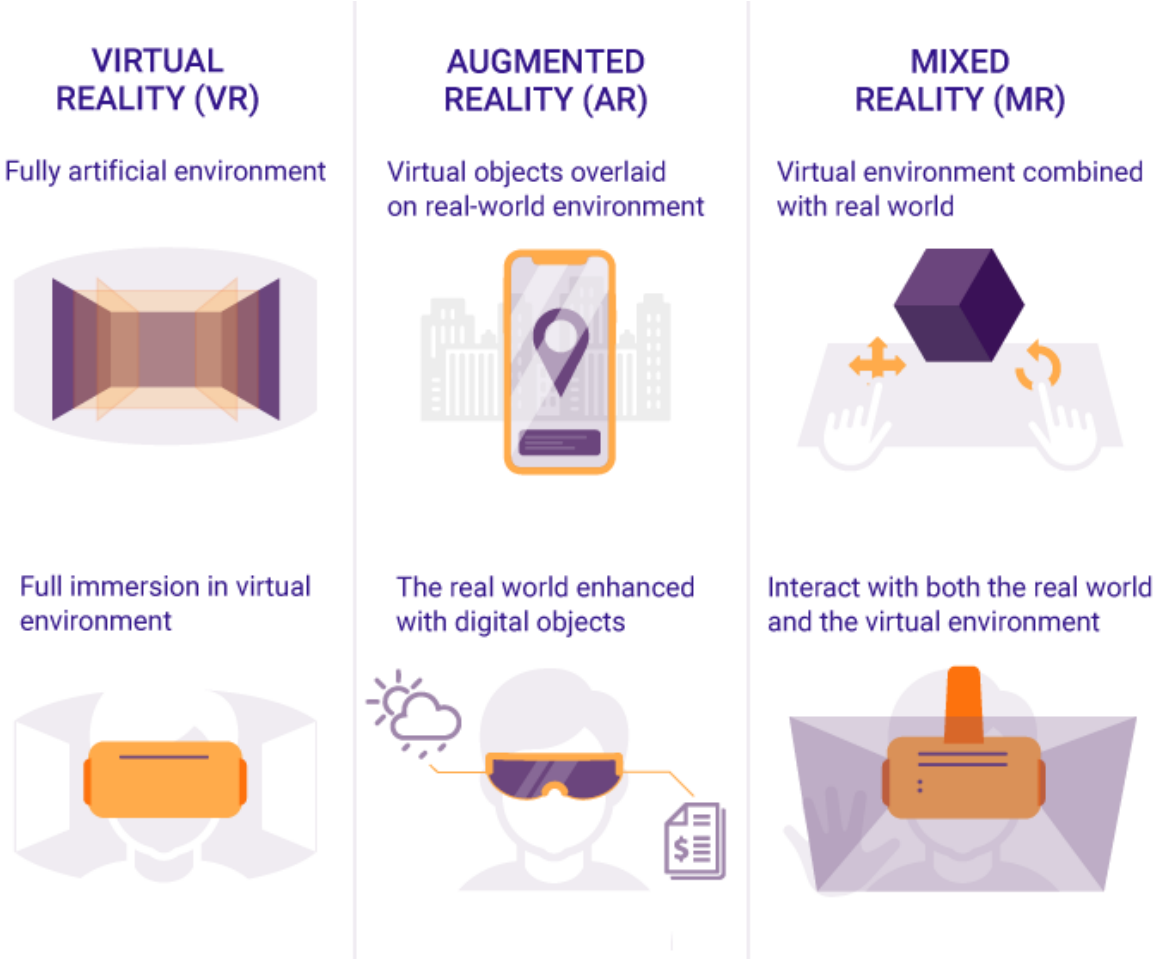
- Also known as decentralized apps (dApps)
- Leverages smart contracts and blockchain data to offer transparency, security and giving back control to users over their assets and data (user-centric)
- Many Web3 apps make money by selling NFTs (Non-Fungible Tokens). E.g., Web3 games sell NFTs representing in-game assets like characters, skins, items, or virtual real estate to enable players to have true ownership over digital items.



Use of Artificial Intelligence



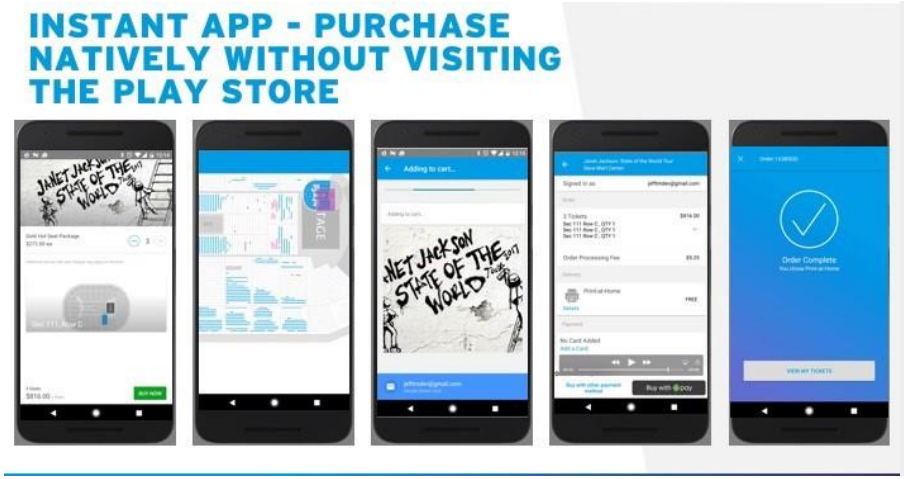
Use of VR & AR

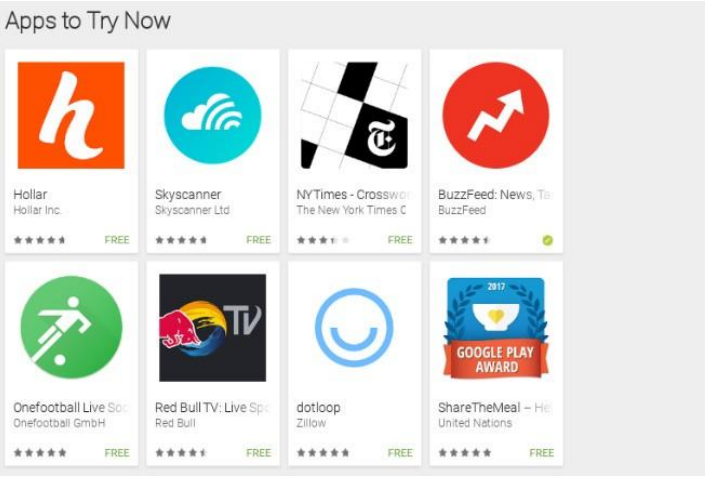


Example: Pokemon Go – according to reports, Pokemon Go has generated a revenue of $1.2 billion and was downloaded approx. 752 million times. Around 5 million active users playing Pokemon Go across the globe.

Instant Apps for Mobile App Development

- An instant app is a small software program that enables end users to test out a portion of a native Android app without installing it on a device.
- Instant apps, although they run like local apps, are native containers with access to a device's hardware.
- Because end users do not install them, instant apps do not take up storage on the device.
- Starting from version 3.0, Android Studio support developing instant apps or transforming full apps into an instant apps.





Mobile Health (mHealth)

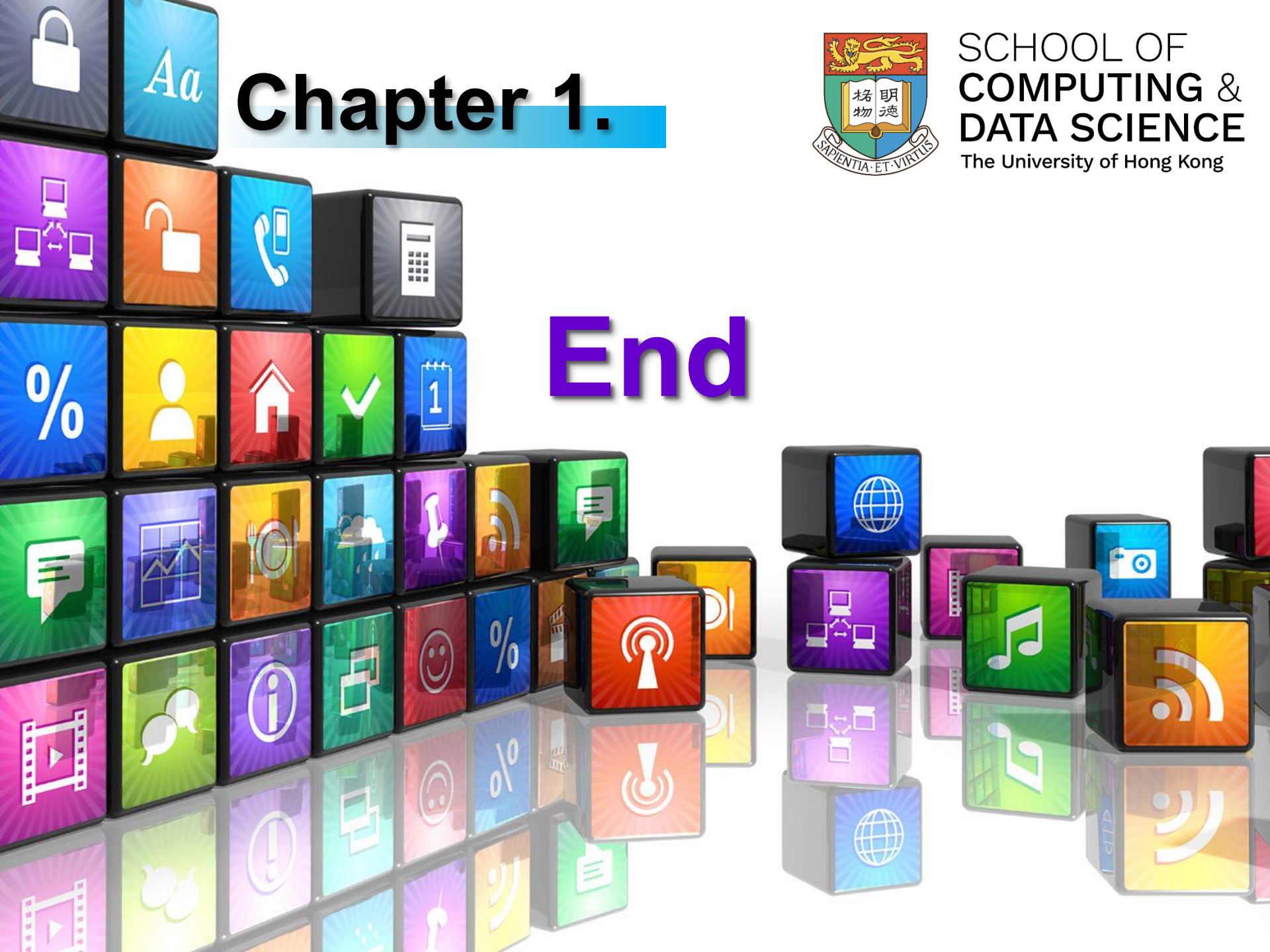
- Use mobile applications to support healthcare service and to provide health information
- Advantages over traditional healthcare approaches:
  - Real-time health service
  - Flexible health service (e.g., telemedicine)
  - Serve more patients
- Applications nowadays:
  - Disease screening and management (e.g., cancers, cardiovascular diseases, diabetes, etc.)
  - Patient self-management (e.g., medication and treatment reminders, body weight, calorie counting, food tracking)
  - Health service provider searching
  - Patient education (e.g., disease symptoms, medications and drugs)
- Feature requirements:
  - Time efficiency (important)
  - User-friendly (important)
  - Simple (important)
  - Funny elements such as games and social media integration (to sustain long-term use)

mHealth Apps for Elderly

- Given that the Hong Kong population is rapidly ageing, older people will become the primary consumers of mHealth apps.
- Examples:
  - Elderly care (e.g., Fall Prevention)
  - Remote or self-monitoring (e.g., chronic conditions such as hypertension)
  - Patient education
  - Finding / booking a healthcare service (e.g., Kowloon Hospital Guide, HA Go)
  - Checking emergency service status
  - Achieving healthy diet (e.g., CookSmart)
  - Remote vital sign monitoring, information exchange, social connection, mental health consultation (in response to the COVID-19 outbreak)

- Problem: Elderly are least likely to accept new technology (e.g., only two-third of older adults aged 65+ owned smartphone and even fewer for older people).
- Challenge: How to attract the elderly to pick up mHealth apps easily and to attract them to use continuously?

Chapter 1.



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